

SatpudaCore

Google Drive Backup -- Complete Setup Guide

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This guide explains how to set up automatic Google Drive backup for every store running SatpudaCore. Backups run silently every 1 hour when internet is connected. The store owner sees nothing.

What	Detail
Backup frequency	Every 1 hour while app is open
On app open	Immediate backup on launch
On app close	Final backup before exit
No internet	Pendrive backup still runs; Drive skipped, retried next hour
Drive error	Warning dialog shown to user
Files kept per day	First 2 + Last 2 of each day; middle ones deleted
Retention period	Files older than 3 years deleted automatically
Pendrive backup	Auto-detected; copies to SatpudaCore_Backup\ on USB drive
Storage cost	Free (15 GB Gmail quota)
Store owner visibility	None -- completely hidden

Phase 1 -- One-Time Google Cloud Setup

NOTE: Do this ONCE on your personal machine. Never repeat unless you create a new project.

Step 1 -- Create a Personal Gmail Account

Use a **personal @gmail.com account** -- not a work or school account.

1. Go to **accounts.google.com** -> Create account -> For personal use
2. Suggested email: **satpudabackupadmin@gmail.com**
3. Save the email and password in your own secure records

WARNING: Do NOT use a Google Workspace / company / school account. They block the required features.

Step 2 -- Create Google Cloud Project

1. Go to **console.cloud.google.com** -- sign in with the Gmail from Step 1
2. Click **Select a project** (top left) -> **New Project**
3. Project name: **SatpudaCore Backup** -> Click **Create**
4. Wait a few seconds -> select the new project from the dropdown at the top

Step 3 -- Enable Google Drive API

1. Left menu -> **APIs & Services** -> **Library**
2. Search: **Google Drive API**
3. Click it -> Click **Enable**

Step 4 -- Configure OAuth Consent Screen

1. Left menu -> **APIs & Services** -> **OAuth consent screen**
2. Select **External** -> Click **Create**
3. App name: **SatpudaCore Backup**
4. User support email: your Gmail address
5. Developer contact email: your Gmail address
6. Click **Save and Continue** through all steps until the last page
7. On the last page click **Back to Dashboard**
8. Scroll down to **Test users** section -> Click **+ Add Users**
9. Enter your Gmail address -> Click **Save**

NOTE: The app stays in Testing mode forever for personal use. No Google verification needed. Up to 100 test users supported.

Step 5 -- Create OAuth2 Client ID and Download JSON

1. Left menu -> **APIs & Services** -> **Credentials**
2. Click **+ Create Credentials** -> **OAuth 2.0 Client ID**
3. Application type: **Desktop app**
4. Name: **SatpudaBackup** -> Click **Create**
5. A dialog appears -> Click **Download JSON**
6. The file downloads with a long auto-generated name like:

```
client_secret_123456789-abcdefghijklmnopqrstuvwxyz.apps.googleusercontent.com.json
```

7. **Rename** this file to exactly:

```
oauth_client.json
```

8. **Move or copy** it to the project root folder (same folder as main.py):

```
d:\satpuda medical store app\satpuda core\mac2\oauth_client.json
```

WARNING: The file **MUST** be named exactly `oauth_client.json` and placed in the project root folder. If the name is wrong or the location is wrong, the next step will fail.

Step 6 -- Install Required Python Packages

Open Command Prompt and run:

```
pip install google-api-python-client google-auth google-auth-oauthlib
```

Wait for installation to complete. You should see: Successfully installed ...

Step 7 -- Generate Encrypted Token (Run Once)

Open Command Prompt, navigate to the project folder, and run:

```
cd "d:\satpuda medical store app\satpuda core\mac2"
```

```
python generate_oauth_token.py
```

What happens step by step:

1. A browser window opens automatically
2. Sign in with your Gmail account (the same one from Step 1)
3. You may see a warning screen: **Google hasn't verified this app**
4. Click **Advanced** (bottom left of the warning screen)
5. Click **Go to SatpudaCore Backup (unsafe)**
6. Click **Allow** on the permissions screen
7. Browser shows: **The authentication flow has completed. You may close this window.**
8. Back in Command Prompt you see:

Done! backup_creds.dat saved to: ...\\config\\backup_creds.dat This file contains your encrypted OAuth2 refresh token. Bundle it with every EXE build -- it never expires.

OK: This token never expires as long as you do not revoke access. You never need to run this script again unless you change your Gmail account. backup_creds.dat is automatically bundled in every EXE you build.

Step 8 -- Create Drive Backup Folder

1. Go to **drive.google.com** -- same Gmail account
2. Click **+ New -> Folder**
3. Name it: **SatpudaCore Backups** -> Click **Create**
4. Double-click to open the folder
5. Look at the URL in your browser address bar:

<https://drive.google.com/drive/folders/1BxiMVs0XRA5nFMdKvBdBZjgmUUqptlbs74ogVE>

6. Copy everything after **/folders/** -- that is your **FOLDER_ID**
7. Save this ID -- you use the SAME folder ID for every store activation

NOTE: This is the SAME folder ID for ALL stores forever. Subfolders per store are created automatically inside it.

Phase 2 -- Per Store Setup (At Activation Time)

NOTE: Do this ONCE per store when you visit to activate the software. Takes 30 seconds.

Step 9 -- Write Backup Config on Store's Machine

When you are at the store's machine, open Command Prompt in the project folder:

```
cd "d:\satpuda medical store app\satpuda core\mac2"
```

Run this command -- replace the folder ID and store name with real values:

```
python setup_store_backup.py YOUR_FOLDER_ID "Store Name Here"
```

Example with the actual folder ID:

```
python setup_store_backup.py 1hNbs7Kxe36V41Ohlo0fc7LxIOk7OiKhC "Satpuda Vet Nagpur"
```

You should see:

```
backup_config.dat written successfully. Store : Satpuda Vet Nagpur Folder:
1hNbs7Kxe36V41Ohlo0fc7LxIOk7OiKhC Backups will start silently on next app launch.
```

OK: The store owner sees nothing. Backups start automatically on next app open.

Step 10 -- Verify Backup Works

Test the backup manually before leaving the store:

```
python -c "from core.backup_manager import run_backup_now; run_backup_now()"
```

Then check the log:

```
type config\backup_log.txt
```

Last line should show:

```
INFO Backup OK - SatpudaCore_2026-04-24_10-30.db.gz -> Satpuda Vet Nagpur
```

Also verify on Drive: go to **drive.google.com** -> **SatpudaCore Backups** -> **Store_Satpuda_Vet_Nagpur**
-- the .db.gz file should be there.

Store Record -- Keep This Safe

Store Name	Folder ID (same for all)	Activation Date	Notes
	1hNbs7Kxe36V41Ohlo0fc7LxIOk7OiKhC		
	1hNbs7Kxe36V41Ohlo0fc7LxIOk7OiKhC		
	1hNbs7Kxe36V41Ohlo0fc7LxIOk7OiKhC		
	1hNbs7Kxe36V41Ohlo0fc7LxIOk7OiKhC		
	1hNbs7Kxe36V41Ohlo0fc7LxIOk7OiKhC		

Phase 3 -- How Backup Works in the App

Backup Schedule

Trigger	When	Notes
App opens	Immediately on launch	New file with current timestamp
Every 1 hour	While app is running	New file each time
App closes	Final backup before exit (blocking)	Skipped if last backup was < 5 min ago
Manual	Settings -> Database -> Backup Now button	Skipped if last backup was < 5 min ago

NOTE: Each backup creates a NEW file with its own timestamp -- old files are NEVER overwritten or replaced. Open at 9:00 = SatpudaCore_..._09-00.db.gz. Close at 9:02 = skipped (within 5-min dedup window). Close at 9:10 = SatpudaCore_..._09-10.db.gz (separate file).

Internet Check

Before every backup, the app checks internet by trying to connect to 4 different servers:

8.8.8.8:53 (Google DNS) | 1.1.1.1:53 (Cloudflare) | google.com:80 | google.com:443

If ALL four fail -> backup is silently skipped. No error shown. Retried next hour.

Drive Connection Error

If internet is available but Google Drive connection fails (token issue, network block, etc.) -> a warning dialog is shown to the user:

NOTE: Automatic backup could not connect to Google Drive. Your data is safe locally. Backup will retry in 1 hour.

File Retention -- 4 Files Per Day, 3-Year Limit

After each upload, the app keeps **4 files per day** and deletes the middle ones:

- File 1: First backup of the day (app opens in morning)
- File 2: Second backup of the day (1 hour after open)
- File 3: Second-last backup of the day
- File 4: Last backup of the day (app closes in evening)
- All middle files between File 2 and File 3 are deleted automatically
- **Files older than 3 years are permanently deleted** from Drive and pendrive

Example of what you see on Drive for one store:

```
Store_Satpuda_Vet_Nagpur/ SatpudaCore_2026-04-26_09-00.db.gz <- 1st of day
SatpudaCore_2026-04-26_10-00.db.gz <- 2nd of day SatpudaCore_2026-04-26_17-00.db.gz <-
2nd last of day SatpudaCore_2026-04-26_18-00.db.gz <- last of day
SatpudaCore_2026-04-25_09-00.db.gz ... (4 files per day, kept for 3 years)
```

Storage estimate: 4 files x 500 KB x 365 days x 3 years = ~2.1 GB per store. 15 GB free Gmail storage handles ~7 stores.

Manual Backup Button

Location: **Settings -> Database tab -> Google Drive Backup section -> Backup Now**

Shows result after clicking:

- Backup successful!
- No internet connection.
- Backup not configured. (run setup_store_backup.py first)
- Backup failed. Check backup_log.txt.

Restoring a Backup

1. Log into your Gmail -> drive.google.com
2. Open SatpudaCore Backups -> Store folder
3. Download the latest .db.gz file
4. Open Command Prompt and run:

```
python -c "import gzip, shutil;
open('veterinary.db', 'wb').write(gzip.open('SatpudaCore_xxx.db.gz', 'rb').read())"
```

5. Open veterinary.db with **DB Browser for SQLite** -- all data is there

Phase 3B -- Backup File Naming, Full Restore & Pendrive Backup

How to Read the Backup File Name

Every backup file follows this exact format:

SatpudaCore_YYYY-MM-DD_HH-MM.db.gz

Example: **SatpudaCore_2026-04-24_10-30.db.gz**

Part	Meaning	Example
SatpudaCore	Fixed prefix -- always present	SatpudaCore
YYYY-MM-DD	Date backup was taken (Year-Month-Day)	2026-04-24 = 24 April 2026
HH-MM	Time backup was taken (24-hour clock)	10-30 = 10:30 AM
.db.gz	Gzip-compressed SQLite database file	--

NOTE: The date and time in the filename are taken from the store machine's local clock at the moment the backup ran.

Step-by-Step: Download from Google Drive

1. Open **drive.google.com** in your browser and sign in with the backup Gmail
2. Navigate to **SatpudaCore Backups** folder
3. Open the subfolder named **Store_<StoreName>** (e.g. Store_Satpuda_Vet_Nagpur)
4. Find the file you want -- pick the most recent, or a specific date if needed
5. Right-click the file -> **Download**
6. The .db.gz file saves to your Downloads folder

Step-by-Step: Extract the .db.gz File

The file is gzip-compressed. You must extract it to get **veterinary.db**.

Option A -- 7-Zip (recommended, free):

1. Install 7-Zip from <https://www.7-zip.org> if not already installed
2. Right-click the downloaded .db.gz file
3. Choose 7-Zip -> Extract Here
4. You get a file named **veterinary.db**

Option B -- Windows 11 built-in:

1. Right-click the .db.gz file -> Extract All
2. You get **veterinary.db**

Option C -- Python command (if Python is installed):


```
python -c "import gzip,shutil;
shutil.copyfileobj(gzip.open('SatpudaCore_2026-04-24_10-30.db.gz','rb'),
open('veterinary.db','wb'))"
```

Replace the filename with your actual downloaded file name.

Step-by-Step: Restore the Database

WARNING: Always rename the old database file BEFORE replacing it. Never delete it directly. This lets you go back if something goes wrong.

1. **Close the app completely** -- make sure it is NOT running
2. Go to the folder where **VeterinaryManagementSystem.exe** is located
3. You will see the current **veterinary.db** file there
4. Rename the current file as a safety copy:

```
veterinary.db -> veterinary_old_backup.db
```

5. Copy the extracted **veterinary.db** into that same folder
6. Double-click **VeterinaryManagementSystem.exe** to start the app
7. Verify your data is restored correctly
8. Once confirmed, you can delete veterinary_old_backup.db

Pendrive (USB) Backup -- Automatic

The app automatically detects any connected pendrive (USB drive) and copies the backup to it -- **no setup needed, works even without internet.**

What	Detail
Detection method	Scans drive letters D: to Z: for removable drives (Windows)
Trigger	Every backup cycle (every 1 hour), even without internet
Folder created on pendrive	SatpudaCore_Backup\Store_<StoreName>\
Retention on pendrive	Same as Drive: 3-year limit + first 2 / last 2 per day
Multiple pendrives	First detected removable drive is used
No pendrive connected	Silently skipped, no error

Example folder structure on the pendrive:

```
E:\SatpudaCore_Backup\ Store_Satpuda_Vet_Nagpur\ SatpudaCore_2026-04-24_08-00.db.gz
SatpudaCore_2026-04-24_10-30.db.gz SatpudaCore_2026-04-25_09-00.db.gz
```

To restore from pendrive: Follow the same Extract and Restore steps above. Just pick the .db.gz file directly from the pendrive instead of downloading from Drive.

NOTE: The pendrive must have enough free space. Each backup file is typically under 1 MB (compressed). 4 files/day x 365 days x 3 years = ~4,380 files = ~4 GB max.

Phase 4 -- Building the EXE

Files Bundled in EXE

File	Purpose
config/backup_creds.dat	Encrypted OAuth2 token -- your Gmail credentials
config/backup_config.dat	Written at activation -- store folder ID (in AppData)
config/backup_log.txt	Auto-created silent log -- never shown to user

Build Command

```
build_exe.bat
```

backup_creds.dat is already listed in both .spec files and gets bundled automatically.

Complete Checklist

When	What	Command / Action
Once ever	Create Gmail + Cloud project	console.cloud.google.com
Once ever	Enable Drive API	APIs & Services -> Library
Once ever	Configure OAuth consent screen	Add yourself as test user
Once ever	Create OAuth2 Client ID	Download JSON file
Once ever	Rename JSON file	Rename to oauth_client.json
Once ever	Move JSON to project folder	Same folder as main.py
Once ever	Install packages	pip install google-api-python-client google-auth google-auth-oauthlib
Once ever	Generate token	python generate_oauth_token.py
Once ever	Create Drive folder	drive.google.com -> New Folder
Once ever	Copy folder ID	From Drive URL after /folders/
Per store	Write backup config	python setup_store_backup.py FOLDER_ID "Name"
Per store	Verify backup	Check backup_log.txt and Drive
Each build	Build EXE	build_exe.bat (auto-bundles creds)